

Formula sheet

~~$$PR_i = \frac{P_{it} - P_{i(t-1)}}{P_{i(t-1)}} \quad PR_I = \sum_{i=1}^n \omega_i PR_i = \sum_{i=1}^n \omega_i \left(\frac{P_{it} - P_{i(t-1)}}{P_{i(t-1)}} \right)$$

$$TR_i = \frac{P_i - P_{i(t-1)} + \text{Income}}{P_{i(t-1)}} \quad w_i^P = \frac{P_i}{\sum_{i=1}^N P_i} \quad r^P = \sum_{i=1}^N w_i^P r_i \quad \text{Index}^P = \frac{\sum_{i=1}^N P_i}{d}$$

$$\text{new divisor} = \frac{\text{sum of constituents before split}}{\text{index value before split}}$$

$$w_i^M = \frac{Q_i P_i}{\sum_{i=1}^N Q_i P_i} \quad \omega_i^{\text{float}} = \frac{f_i Q_i P_i}{\sum_{j=1}^N f_j P_j Q_j} \quad r^M = \sum_{i=1}^N w_i^M r_i$$

$$\text{Index}^M = \frac{\sum_{i=1}^N Q_i P_i}{d}$$

$$d = \frac{\sum_{i=1}^N Q_{Bi} P_{Bi}}{\text{Index}^M} \quad w_i^E = \frac{1}{N} \quad r^E = \frac{\sum_{i=1}^N r_i}{N} \quad \text{Index}^E = \frac{\sum_{i=1}^N Q_i P_i}{d}$$

$$d = \frac{\sum_{i=1}^N Q_{Bi}^* P_{Bi}}{\text{Index}_B^E} \quad \omega_i = \frac{F_i}{\sum_{j=1}^n F_j}$$~~

choice under uncertainty + optimal Portfolios (EUT)

$$E(U) = \sum_w U(W)P(W) \quad ARA = \frac{-U''(W)}{U'(W)} \quad RRA = W \frac{-U''(W)}{U'(W)}$$

$$RP = \frac{1}{2} A(w) \cdot \sigma^2 \quad \text{Variance} = \sigma^2 = \sum_{i=1}^n [R_i - E(R_i)]^2 P_i$$

$u(CE) = E(U)$
 $RP = \text{Initial wealth} - CE$

$$\text{Standard Deviation} = \sigma = \sqrt{\sum_{i=1}^n [R_i - E(R_i)]^2 P_i} \quad \text{Cov}_{ij} = E\{[R_i - E(R_i)][R_j - E(R_j)]\}$$

corr = $\rho_{i,j} = \frac{\text{cov}_{i,j}}{\sigma_i \sigma_j}$

$$\sigma_p^2 = \omega_1^2 \sigma_1^2 + 2\omega_1 \omega_2 \rho_{12} \sigma_1 \sigma_2 + \omega_2^2 \sigma_2^2$$

$$= \omega_1^2 \sigma_1^2 + 2\omega_1 \omega_2 \sigma_{12} + \omega_2^2 \sigma_2^2$$

$$U = E(r) - \frac{1}{2} A \sigma^2$$

$$\sigma_p^2 = \omega_1^2 \sigma_1^2 + \omega_2^2 \sigma_2^2 + \omega_3^2 \sigma_3^2 + 2\omega_1 \omega_2 \sigma_{1,2} + 2\omega_1 \omega_3 \sigma_{1,3} + 2\omega_2 \omega_3 \sigma_{2,3}$$

SML(β) = $S = \frac{E(r_p) - r_f}{\sigma_p}$

$$w_1 = \frac{R_1 \sigma_2^2 - R_2 \sigma_{1,2}}{R_1 \sigma_2^2 + R_2 \sigma_1^2 - (R_1 + R_2) \sigma_{1,2}}$$

$$y = \frac{E(r_p) - r_f}{A \sigma_p^2}$$

$$\omega_1 = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$$

$$\omega_2 = 1 - \omega_1$$

CAPM

$$\omega_1 = \frac{\sigma_2^2 - \rho_{12} \sigma_1 \sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12} \sigma_1 \sigma_2}$$

$$r_i = r_f + \beta(r_m - r_f), \text{ where } \beta_i = \frac{\sigma_{im}}{\sigma_m^2}$$

CAPM

APT single index model

$$R_i = \alpha_i + \beta_i R_M + e_i$$

$$\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma^2(e_i)$$

$$\text{Cov}(r_i, r_j) = \text{Cov}(\beta_i m + e_i, \beta_j m + e_j) = \beta_i \beta_j \sigma_m^2$$

APT

$$R_{it} = a_i + \sum_{j=1}^k b_{ij} F_{jt} + \varepsilon_{it} \quad ER_{it} = \lambda_0 + \sum_{j=1}^k b_{ij} \lambda_j$$

APT

$$E(R_i) = \alpha_i + \beta_{iM} E(R_M) + \beta_{iSMB} E(R_{SMB}) + \beta_{iHML} E(R_{HMLt}) + e_{it}$$

$$E(R_i) = \alpha_i + \beta_{iM} E(R_M) + \beta_{iSMB} E(R_{SMB}) + \beta_{iHML} E(R_{HMLt}) + \beta_{iWML} E(R_{WMLt}) + e_{it}$$

$$E(R_i) = \alpha_i + \beta_{iM} E(R_M) + \beta_{iSMB} E(R_{SMB}) + \beta_{iHML} E(R_{HMLt}) + \beta_{iCMA} E(R_{CMAt}) + \beta_{iRMW} E(R_{RMWt}) + e_{it}$$

Tracking Error

return d. (Frenkel)

$$\Delta_t = \sum_{i=1}^N w_i R_{it} - R_{bt} = R_{pt} - R_{bt} \quad \sigma_{\Delta}^2 = \frac{\sum_{t=1}^T (\Delta_t - \bar{\Delta})^2}{(T-1)}$$

$$\sigma_{\Delta} = \sqrt{\sigma_{\Delta}^2} = \text{periodic tracking error} \quad \text{Annualized } TE = \sigma_{\Delta} \sqrt{P}$$

Treynor-Black

$$w_i^0 = \frac{\alpha_i}{\sigma^2(e_i)}$$

$$w_i = \frac{w_i^0}{\sum_{i=1}^n w_i^0}$$

$$\alpha_A = \sum_{i=1}^n w_i \alpha_i$$

$$\sigma^2(e_A) = \sum_{i=1}^n w_i^2 \sigma^2(e_i)$$

$$w_A^0 = \left[\frac{\alpha_A / \sigma^2(e_A)}{RP_M / \sigma_M^2} \right]$$

$$\beta_A = \sum_{i=1}^n w_i \beta_i$$

$$w_A^* = \frac{w_A^0}{1 + (1 - \beta_A) w_A^0}$$

$$E(R_p) = (w_M^* + w_A^* \beta_A) RP_M + w_A^* \alpha_A$$

$$\sigma_p^2 = \beta_p^2 \sigma_m^2 + w_A^{*2} \sigma^2(e_A)$$

$$S_p^2 = S_M^2 + \left[\frac{\alpha_A}{\sigma(e_A)} \right]^2$$

$$\text{or } S_p = \frac{E(R_p)}{\sigma_p}$$

$$\sigma_A = \beta_A \sigma_m + \sigma^2(e_A) \quad \text{total value of the securities sold} \\ \text{Fund turnover} = \frac{\text{average value of the assets managed by the fund.}}$$

$$\text{Tax cost ratio} = \left[1 - \frac{(1 + TAR)}{(1 + PTR)} \right] \times 100,$$

Treynor

$$t_i = \frac{E(R_i) - E(R_f)}{\beta_i}$$

$$S_p = \frac{E(R_p) - E(R_f)}{\sigma_p} \quad \text{Sharpe}$$

Jansen's alpha

$$\alpha_p = R_{p,t} - \left[R_{f,t} + \beta_p (R_{m,t} - R_{f,t}) \right] \quad M_{p.}^2 = r_p + S_p * (\sigma_M - \sigma_p) \quad m^2$$

Sortino

$$ST_i = \frac{E(R_i) - \tau}{DR_i}$$

$$= \sqrt{\frac{1}{n} \sum_{R < E(R)} (R_{it} - E(R))^2}$$

Info ratio

$$IR_p = \frac{\alpha_p}{\sigma(e_p)}$$

$$MDD = \left| \frac{(\text{Trough Value} - \text{Peak Value})}{\text{Peak Value}} \right|$$

* abs. value

$$\beta_F = \frac{\sigma_p}{\sigma_b} \quad \text{Calmar}_p = \frac{r_p}{MDD}$$

Diversification

$$\text{Net Selectivity} = \underbrace{(r_p - \rho(r_0 - r_f) - r_f)}_{\text{Selectivity}} + \underbrace{(\rho_F - \rho)(r_0 - r_f)}_{\text{Diversification}}$$

$$= \alpha - d$$

